

Intel® Infrastructure Power Manager (IPM) Operator

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Introduction

High performance containerized workloads are currently readily deployed in an orchestrated fashion within a Kubernetes environment, such as Red Hat OpenShift Container Platform (RHOCP). However, applications with critical SLA requirements typically favor performance over efficient power consumption. The ideal scenario is to be able to achieve both performance and power optimization. Intel® Infrastructure Power Manager (IPM) is a comprehensive software solution for automatically managing and optimizing power settings across diverse data center environments. It provides granular control of processor power settings, whether provisioning infrastructure post-boot or dynamically reducing power consumption at runtime.

In other words, IPM saves power while maintaining system performance under real-world operating conditions. It also acts as a single, platform-independent controller for all power management-related features in a server that higher-level management software can query or control at runtime.

This document focuses on configuring and installing the Intel® Infrastructure Power Manager Operator in a Red Hat OpenShift environment.

This paper includes the following sections:

- The Overview describes the cluster topology, the hardware along with the software configuration.
- Phase-1 describes pre-requisites for the RHOCP cluster prior to building and deploying the IPM agent along with the proxy workload.
- Phase-2 Installing the IPM operator.
- Phase-3 describes the step-by-step process for validating the IPM operator deployment.

Note: The IPM is designed for use on Intel® CPU's only and works across a range of Intel® Xeon® and Intel® Core™ processors. For access to more comprehensive information and documentation on the IPM, please contact your local Intel sales representative. Please review the [IPM license](#) for more information on commercial use.

Overview

The scope of this document is to walk through, step-by-step, to onboard the Intel® Infrastructure Power Manager operator.

This document assumes that Red Hat OpenShift has been installed on the cluster. The Red Hat OpenShift installation can be achieved in many ways as described in Red Hat OpenShift installation [documentation](#). For general overview of the operator please refer to operator related [document](#).

It is divided into three phases namely:

Phase 1: Prepare the deployed OpenShift cluster by ensuring IPM operator's prerequisites are met and configuring the BIOS.

Phase 2: Deploy the Intel® Infrastructure Power Manager operator on top of the cluster.

Phase 3: Validate the Intel® Infrastructure Power Manager deployment

i. References and Resources

Here is a list of references and resources used within this user guide. Resources with an Intel® document ID may be obtained from the [Intel® Resource & Documentation Center \(RDC\)](#) by searching for the corresponding document ID.

Note that RDC has restricted access for customers under NDA (Non-Disclosure Agreement) with Intel®. To find out how to obtain access to RDC please contact your local Intel® sales representative.

Document	Document ID/Link
Intel® Infrastructure Power Manager – Technical Integration Guide	861477
Red Hat OpenShift Container Platform documentation links	779745

ii. Hardware Configuration

Intel® Infrastructure Power Manager has been validated on servers based on 2nd to 6th Generation Intel® Xeon® Scalable processors and on 12th to 14th Generation Intel® Core™ processors.

iii. Software configuration

List of software components used in this user guide:

Category	Components	Details
Firmware	BIOS firmware	Please use the latest published OEM firmware
OS	Red Hat OpenShift	This operator is certified on 4.16, 4.18
Red Hat OpenShift Operator	Intel® Infrastructure Power Manager Operator	IPM version 25.07

Phase-1: Prerequisites for the Intel® Infrastructure Power Manager Operator

i. Prerequisite for IPM Operator

Tuned

The following Tuned profile must be applied to the cluster to ensure that Tuned does not manage any settings that are managed by IPM. The cluster must have only one entity managing power related settings, and if the intention is to deploy IPM, this entity will be IPM only. If this profile is deployed on the cluster, any settings configured in Tuned's CPU or Uncore plugins will be ignored, however other Tuned plugins can still be used.

1. Save the code block provided below as *disable-cpu-uncore.yaml*

```
apiVersion: tuned.openshift.io/v1
kind: Tuned
metadata:
  name: custom-disable-cpu-uncore-plugin
  namespace: openshift-cluster-node-tuning-operator
spec:
  profile:
    - name: disable-cpu-uncore-profile
      data: |
        [main]
        summary=Disable CPU and Uncore plugins for tuned

        [cpu]
        enabled=false

        [uncore]
        enabled=false

  recommend:
    - profile: disable-cpu-uncore-profile
      priority: 0
```

2. Apply the profile:

```
oc apply -f disable-cpu-uncore.yaml
```

Annotations

The following annotations cannot be used with IPM as IPM already manages the configuration parameters associated with these annotations.

```
"cpu-freq-governor.crio.io" / CPUCStatesAnnotation
"cpu-c-states.crio.io" / CPUFreqGovernorAnnotation
```

ii. Validation of BIOS settings

In general, for IPM to automatically adjust the P-state settings from the Operating System (OS):

- Intel® SpeedStep® (Pstates) Technology should be set to "Enable".
- Processor C6 should be set to "Enable".
- Package C State should be set to "C6 Non-retention".
- The Energy Performance Bias Power Performance Tuning option needs to be configured to allow OS control.

Note: IPM does not require that the intel_pstate driver be explicitly loaded by the operating system. In this case, if the grub cmdline includes the "intel_pstate=disable" option, the acpi-cpufreq driver will be used to adjust the P-state settings. The intel_pstate=disable option is included when perPodPowerManagement is set to False in the PerformanceProfile.

Phase-2: Install the Intel® Infrastructure Power Manager Operator

- On logging into the cluster, the following overview is presented

The screenshot displays the Red Hat OpenShift Overview page. The left sidebar contains navigation links for Administrator, Home, Operators, Workloads, Networking, Storage, Builds, Pipelines, Observe, Compute, User Management, and Administration. The main content area is titled 'Overview' and includes sections for 'Getting started resources', 'Details', 'Status', 'Cluster utilization', and 'Activity'.

Details: Cluster API address, Cluster ID, Infrastructure provider, OpenShift version, Service Level Agreement (SLA), Self-support, Update channel, and Control plane high availability.

Status: Cluster, Control Plane, Operators, Insights, and Dynamic Plugins. Alerts include 'ClusterOperatorDegraded', 'AlertmanagerReceiverNotConfigured', and 'ClusterOperatorDegraded'.

Cluster utilization: A table showing resource usage for CPU, Memory, and Filesystem. CPU usage is 93.77% (93.77/100), Memory usage is 48.37 GB (48.37/50 GB), and Filesystem usage is 371.2 GB (371.2/400 GB).

Activity: A list of recent events, including 'Stopping container registry-server', 'Started container registry-server', 'Created container: registry-server', 'Successfully pulled image', 'Created container: extract-content', 'Pulling image', 'Successfully assigned openshift-marketplace/community...', 'Add with [192.168.1.2/3] from own-kubernetes', 'Created container: extract-utilities', 'Container image "quay.io/openshift-release-dev/ocp-v4.0-art...", 'Started container extract-utilities', 'Pulling image "registry.redhat.io/redhat/community-ope...', 'Container image "quay.io/openshift-release-dev/ocp-v4.0-art...', 'storageclass.storage.k8s.io "local-cd" not found', and 'storageclass.storage.k8s.io "local-cd" not found'.

- On the left-hand side navigate to:

- Operators -> Operator Hub

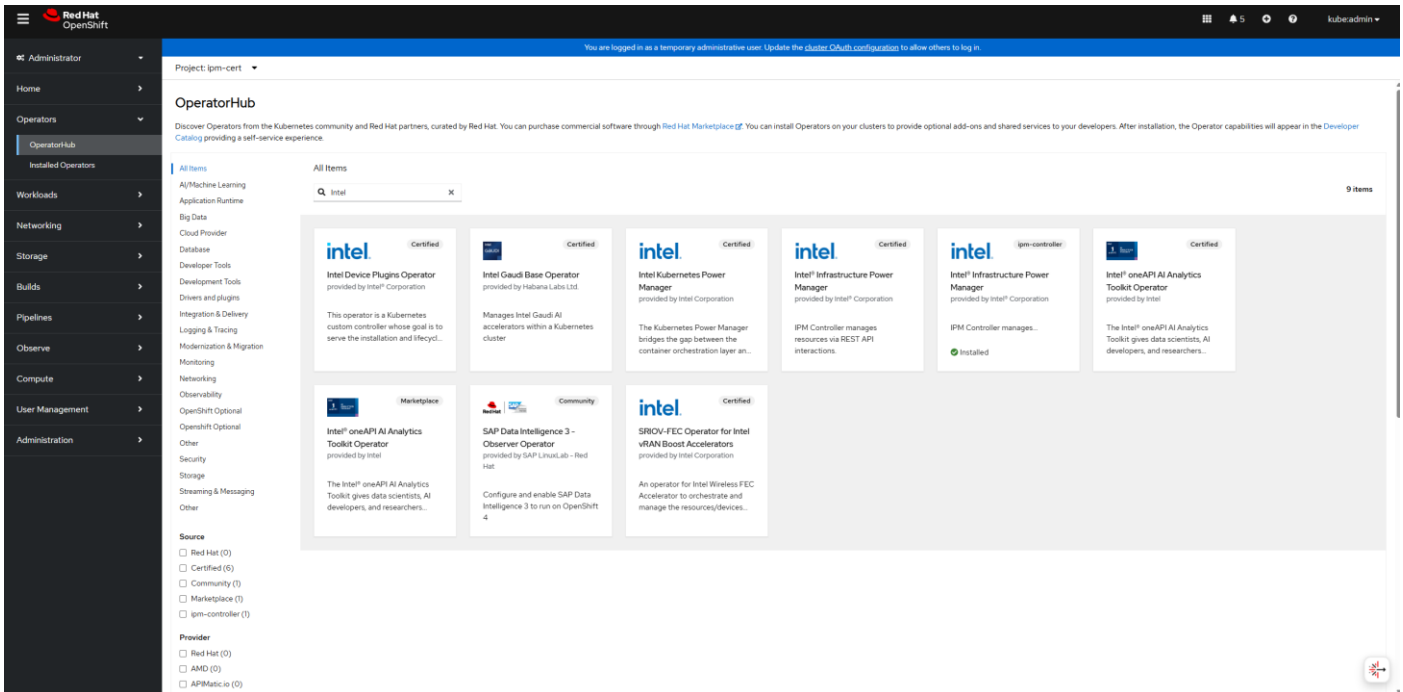
The screenshot displays the Red Hat OpenShift Operator Hub page. The left sidebar contains navigation links for Administrator, Home, Operators, Workloads, Networking, Storage, Builds, Pipelines, Observe, Compute, User Management, and Administration. The main content area is titled 'OperatorHub' and includes a search bar, a list of operators, and a 'Source' section.

Source: Red Hat (136), Certified (279), Community (274), Marketplace (44), and ipm-controller (1).

Operator Grid: A grid of operators available for installation, including 3scale API Management, ABot, AccuKnox Operator, Advanced Cluster Security for Kubernetes, Aerospike Kubernetes Operator, and others.

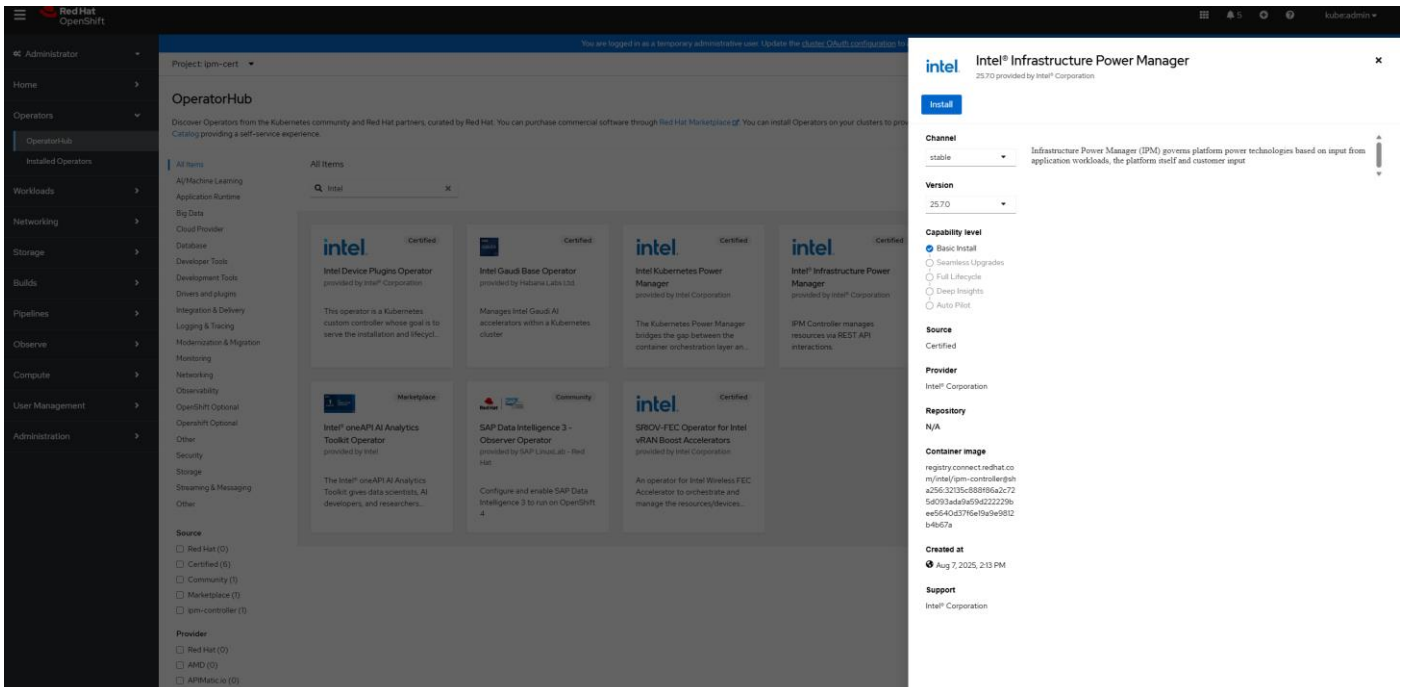
3. And in the search bar type:

1. Intel
2. Which presents a list of all the RHOCIP Intel operators

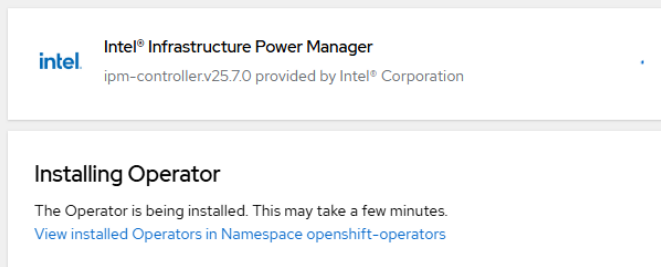


4. Click on:

1. Intel® Infrastructure Power Manger and click install



5. The Intel® Infrastructure Power Manger Operator will be installed by OCP



The screenshot shows the OpenShift console interface for the Intel Infrastructure Power Manager Operator. It features the Intel logo, the operator name, its version, and a progress indicator. Below this, a section titled 'Installing Operator' provides a status message and a link to view installed operators.

Operator	Version	Provided by	Progress
intel	Intel® Infrastructure Power Manager	ipm-controller.v25.7.0 provided by Intel® Corporation	Progress bar

Installing Operator

The Operator is being installed. This may take a few minutes.

[View installed Operators in Namespace openshift-operators](#)

Phase-3: Operator Usability

Once the operator is installed navigate to:

Operators -> Installed Operators -> Intel® Infrastructure Power Manager

The user will be able to see the following details:

The screenshot shows the 'Operator details' page for the 'Intel® Infrastructure Power Manager' operator. The page includes a 'Details' tab, a 'YAML' tab, and a 'Subscription' tab. The 'Details' tab is active, showing the operator's name, version (25.7.0), and provider (Intel® Corporation). It also displays the 'Created at' date (Aug 28, 2025, 2:18 PM) and the 'Links' section with the IPM URL. The 'Description' section states that the IPM governs platform power technologies based on input from application workloads, the platform itself, and customer input. The 'ClusterServiceVersion details' section is also visible.

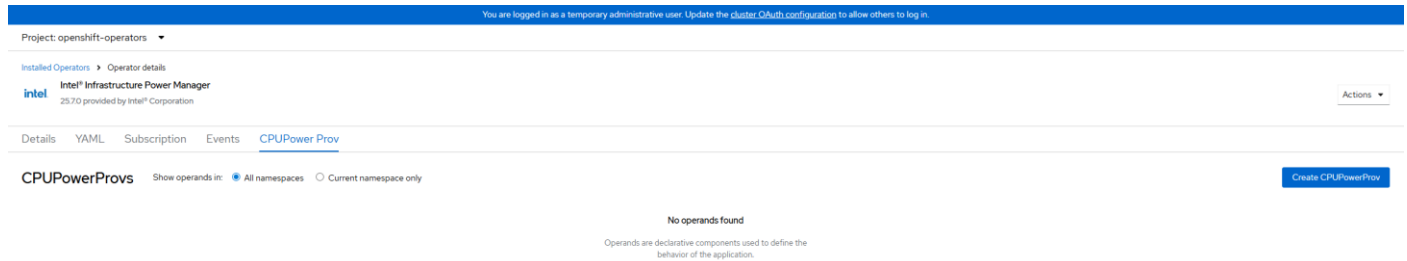
Edits to the YAML for the operator can be made here, this may be required for changing log levels or enabling the idle service for example

The screenshot shows the 'YAML' tab of the operator details page. It displays the full YAML configuration for the operator, including the 'apiVersion', 'kind', 'metadata', 'spec', and 'status' sections. The 'spec' section is expanded, showing the 'deployment' configuration with the 'apiVersion', 'kind', 'metadata', and 'spec' fields. The 'spec' field is further expanded, showing the 'containers' and 'securityContext' fields. The 'containers' field is expanded, showing the 'name', 'image', 'command', and 'args' fields. The 'securityContext' field is expanded, showing the 'privileged' and 'capabilities' fields. The 'status' section is also visible, showing the 'phase' and 'conditions' fields.

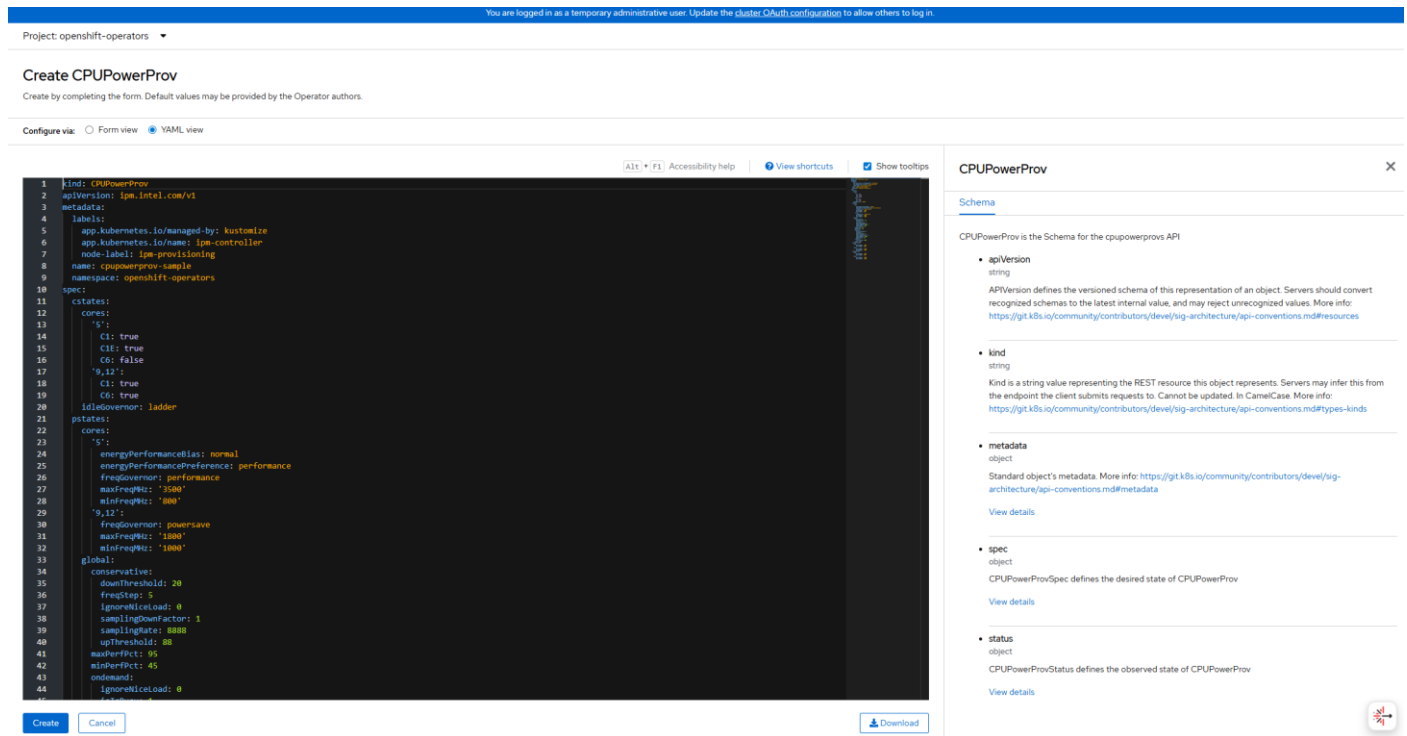
i. Provisioning CPUPower CRD

Power provisioning settings such as C-States, P-States, Uncore, etc. may be deployed to the node or cluster using the CPUPower CRD.

1. Navigate to CPUPower Prov



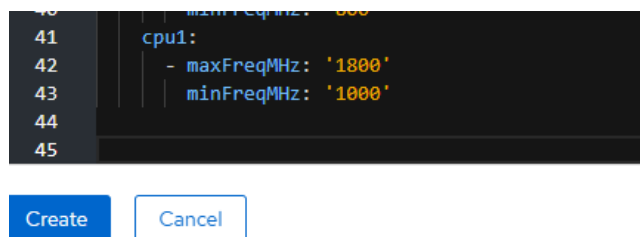
2. Click on create



The recommendation is to use the YAML view to configure the CRD as the autogenerated UI view may not suit the IPM Operator CRD

A default template is provided. It may be necessary to edit some of these settings depending on the system

3. Click on "create" bottom left



4. The CRD can be seen to have been created

You are logged in as a temporary administrative user. Update the cluster `OAuth` configuration to allow others to log in.

Project: openshift-operators ▾

Installed Operators ▸ Operator details

intel Intel® Infrastructure Power Manager
25.70 provided by Intel® Corporation

Details YAML Subscription Events **CPUPowerProv**

CPUPowerProvs Show operands in: ☒ All namespaces ☐ Current namespace only Create CPUPowerProv

Name ▾ Search by name... /

Name	Kind	Namespace	Status	Labels	Last updated
CRD cpupowerprov-sample	CPUPowerProv	openshift-operators	Condition: ✔ Succeeded	app.kubernetes.io/managed-by=kustomize app.kubernetes.io/name=ipm-controller node-label=ipm-provisioning	Aug 29, 2025, 11:14 AM

The CRD has been created. The Status should be Condition Succeeded with a green check mark.

If it has failed, this is probably due to certain provisioning settings not being available. Recheck the YAML and redeploy if necessary

ii. Power Managing Polling Applications

One of the key features of IPM is being able to power manage polling applications that are built using either the Data Plane Development Kit (DPDK) or Vector Packet Processing (VPP) projects. This section will describe from a high level how this can be achieved with the IPM operator for a DPDK application, however more comprehensive information and documentation on this feature can be found in our [Technical Integration Guide \(User Guide\)](#) which requires a CNDA agreement with Intel®. For access, please contact your local Intel® sales representative.

- IPM requires telemetry from a polling application to determine the true busyness of each core running the application. To get this telemetry the application needs to be patched with a patchset from [this directory](#). Patches are available for DPDK 20.11, 21.11, 22.11, 23.11 and 24.11 and VPP 20.09, 21.01, 22.02, 23.02 and 24.02 currently.
- For IPM to pick up the telemetry from the application you will need to make some changes to its podspec also
 - The pod must be labelled so that we know which pods to monitor, the label `ipm-monitored: "true"` must be added to the metadata for the pod.
 - To access the telemetry for DPDK for example the DPDK directory (`/var/run/dpdk`) must be exposed to the host using volume mounts and a volume.
 - In order for IPM to know what telemetry we will monitor for each pod we need to know what telemetry is being used so a configmap with the following data items must be added to the pod (DPDK example below)
 - `telemetry_prefix: /var/run/dpdk`
 - `telemetry_suffix: dpdk_telemetry.v2`
 - Each application must have a unique name so we will use the UUID of the pod for this. The way we recommend doing this is an environment variable in the podspec and passing this PREFIX value as the file prefix for example in DPDK this will name the application using the UUID.
 - Add the environmental variable to the podspec as shown below

```
env:
  - name: PREFIX
    valueFrom:
      fieldRef:
        fieldPath: metadata.uid
```

- The prefix can then be used in the DPDK command arguments as follows

```
./DPDK_APP --file-prefix=${PREFIX}
```

IPM should now automatically detect and power manage your application for more information please refer to our [Technical Integration Guide \(User Guide\)](#)

iii. Enable Logging

By default, logging is set to a minimum level. However, for debugging or test purposes it may be desirable to turn on logging. This is achieved as follows:

1. Edit the YAML for the IPM operator that is installed on the cluster
2. Use Find (CTRL+f) to find the phrase "--zap-log-level=error"
3. This will bring the user to where the log level for the controller is set. The available logs levels are **error**, **info** and **debug** in increasing verbosity. This can be changed to your desired level e.g. "--zap-log-level=info". Log level **info** is recommended for a testing scenario
4. 8 lines below this you there is a line entitled: "-- none"

```
- command:
  - /opt/ipm-agent/ipm-agent
  - '-c'
  - /opt/ipm-agent/config/config.json
  - '-p'
  - '32100'
  - '-l'
  - none
```

This is where the log level for the agent is set, the available logs levels are **none**, **critical**, **error**, **warning**, **info**, **verbose** and **debug** in increasing verbosity. This can be changed to your desired level e.g. "-- info". Log level **info** is recommended for a testing scenario

5. Once the changes are made, save the YAML and reload it to enable

iv. Enabling The Idle Service

The Idle Service feature of IPM dynamically provisions unallocated cores, allocated cores, and uncore of the OCP node with different powersave and performance profiles based on the node's resource usage.

The following steps are needed to enable the idle service on the cluster

1. Edit the YAML for the IPM operator that is installed on the cluster
2. Add --auto-idle-power to the end of the watcher command after the --p 32100. This would make the full watcher command in the YAML:

```
until curl -X GET -sf -o /dev/null http://localhost:32100/global/status --http2-prior-knowledge;
do echo "Waiting for agent . . ."; sleep 2; done; /opt/ipm-watcher/ipm-watcher --p 32100
--auto-idle-power
```

3. Add the following to the watcher container section this is needed to allow the watcher permissions to access the kubelet.sock socket

```
securityContext:
  privileged: true
```

4. Under volume mounts for the watcher container add the following, we need to give the watcher access to the idle services config and the kubelet.sock socket that we will add to the volumes in a separate step

```
- mountPath: /opt/ipm-watcher/config/
  name: ipm-watcher-auto-idle-config
  readOnly: true
- mountPath: /var/lib/kubelet/pod-resources
  name: kubesock
```

This will make the full volume mounts section for the watcher into the following:

```
volumeMounts:
- mountPath: /etc/certs/ipm
  name: certs
  readOnly: true
- mountPath: /opt/ipm-watcher/config/
  name: ipm-watcher-auto-idle-config
  readOnly: true
- mountPath: /var/lib/kubelet/pod-resources
  name: kubesock
```

5. Add the following volumes to the IPM pod

```
- configMap:
  name: ipm-watcher-auto-idle-config
name: ipm-watcher-auto-idle-config
- hostPath:
  path: /var/lib/kubelet/pod-resources
  name: kubesock
```

This will make the full volumes section into the following

```
volumes:
- hostPath:
  path: /sys/devices
  type: Directory
  name: host-devices
- hostPath:
  path: /var/run/dpdk
  type: DirectoryOrCreate
  name: host-run
- hostPath:
  path: /var/run/vpp
  type: DirectoryOrCreate
  name: host-run-vpp
- hostPath:
  path: /dev/cpu
  type: Directory
  name: cpu-msr
- configMap:
  name: ipm-controller-ipm-agent-config
  name: ipm-agent-config
- hostPath:
  path: /etc/certs/ipm/
  name: certs
- emptyDir: {}
  name: heartbeat
- configMap:
  name: ipm-watcher-auto-idle-config
  name: ipm-watcher-auto-idle-config
- hostPath:
  path: /var/lib/kubelet/pod-resources
  name: kubesock
```

6. Then create the configmap for the idle config in a YAML file and apply it to OCP the provided YAML file is just an example and can be changed to suite your cluster. Ensure to change the namespace to the namespace that you installed the IPM operator into or the config will not be accessible by the IPM pod

```
oc apply -f idle-config.yaml

apiVersion: v1
```

```

kind: ConfigMap
metadata:
  name: ipm-watcher-auto-idle-config
  namespace: ipm-test
data:
  auto-idle-save-policy.yaml: |
    corePowerPolicy:
      allocatedCores:
        cstates:
          C1: true
          C1E: true
          C6: false
        pstates:
          minFreqMHZ: "1100"
          maxFreqMHZ: "1800"
          freqGovernor: "performance"
          energyPerformancePreference: "performance"
          energyPerformanceBias: "performance"
      unallocatedCoresPerformance:
        cstates:
          C1: true
          C1E: true
          C6: false
        pstates:
          minFreqMHZ: "1100"
          maxFreqMHZ: "1800"
          freqGovernor: "performance"
          energyPerformancePreference: "performance"
          energyPerformanceBias: "performance"
      unallocatedCoresPowersave:
        cstates:
          C1: true
          C1E: true
          C6: true
        pstates:
          minFreqMHZ: "1000"
          maxFreqMHZ: "1000"
          freqGovernor: "powersave"
          energyPerformancePreference: "power"
          energyPerformanceBias: "normal"
      unallocatedCoresRules:
        excludeNamespaces:
          - kube-system
          - default
        podCountThreshold: 5
        percentCPUReqThreshold: 15
    uncorePowerPolicy:
      idlePowersave:
        computeMinFreqMHZ: "800"
        computeMaxFreqMHZ: "1100"
      activePerformance:
        computeMinFreqMHZ: "1100"
        computeMaxFreqMHZ: "1800"
      rules:
        excludeNamespaces:
          - kube-system
          - default
        podCountThreshold: 5
        percentCPUReqThreshold: 5

```

Summary

Intel® Infrastructure Power Manager is a comprehensive software solution for automatically managing and optimizing power settings across the compute infrastructure. Some of the key benefits are:

- **Dynamic Power Efficiency:** Automatically adjusts CPU power consumption and performance levels in real-time, for active and idle cores, aligning with current workload demands to maximize energy efficiency.
- **Performance-Oriented:** Maintains Key Performance Indicators (KPIs) while simultaneously delivering power savings.
- **Comprehensive Power Provisioning:** Offers fine-grained control over CPU operating frequencies and power states, allowing for tailored power management configurations.

To help scale the use of the IPM across Red Hat OpenShift clusters, Intel® is providing an operator to ease the deployment of the software for the end user. This guide provides details on how to obtain and install the IPM operator.

IPM is designed for use on Intel® CPU's only and works across a range of Intel® Xeon® and Intel® Core™ processors. IPM covers power saving for all scenarios including polling apps, non-polling apps and idle use cases. More comprehensive information and documentation on IPM can be found on [Technical Integration Guide \(User Guide\)](#) which requires CNDA agreement with Intel®. For access, please contact your local Intel® sales representative. Please review the [IPM license](#) for more information on commercial use.